

Assignment 2 – Task 1

Optimizer Performance on Non-Convex Functions

Large Language Models – Assignment 2

1. Overview

The aim of this task was to compare five optimization methods on two non-convex functions: the Rosenbrock function and a one-dimensional simplified Ackley function. The optimizers were tested with learning rates 0.01, 0.05 and 0.1. The implementation was done from scratch in Python, and the experiments recorded the final objective value, number of iterations, execution time and convergence status.

The main result was that Adam was the most reliable method for the Rosenbrock function and reached the global minimum with all three tested learning rates. For the Ackley function, the starting point had a strong effect. From the main starting point $x = 2.3$, the optimizers mostly settled at local minima instead of the global minimum $x = 0$. This was a useful example of why convergence of an optimizer does not always mean that the global optimum has been found.

2. Objectives

- Implement the Rosenbrock and simplified 1D Ackley functions and their gradients.
- Implement Gradient Descent, SGD with Momentum, RMSprop, Adagrad and Adam from scratch.
- Compare the optimizers at learning rates 0.01, 0.05 and 0.1.
- Study convergence behaviour and record the final objective value, iterations and time.
- Check whether the experimental behaviour agrees with the expected effect of learning rate and optimizer design.
- Use the results to identify which methods performed best for the two functions.

3. Functions Used

3.1 Rosenbrock Function

The function used was:

$$f(x, y) = (1 - x)^2 + 100(y - x^2)^2$$

Its global minimum is at $(x, y) = (1, 1)$, where $f(x, y) = 0$. The starting point used for the main experiment was $(-1.5, 2.0)$.

3.2 Simplified 1D Ackley Function

$$f(x) = -20 \exp(-0.2\sqrt{(x^2)}) - \exp(\cos(2\pi x)) + 20 + e$$

The implementation explicitly set $f(0) = 0$ to handle the point $x = 0$ safely. The global minimum is $x = 0$ with $f(x) = 0$. The main experiment started from $x = 2.3$.

3.3 Visualization of the Objective Functions

The notebook also visualized both objective functions before running the optimizers. These plots help show the shape of the search space and explain why the functions are challenging.

Rosenbrock Function

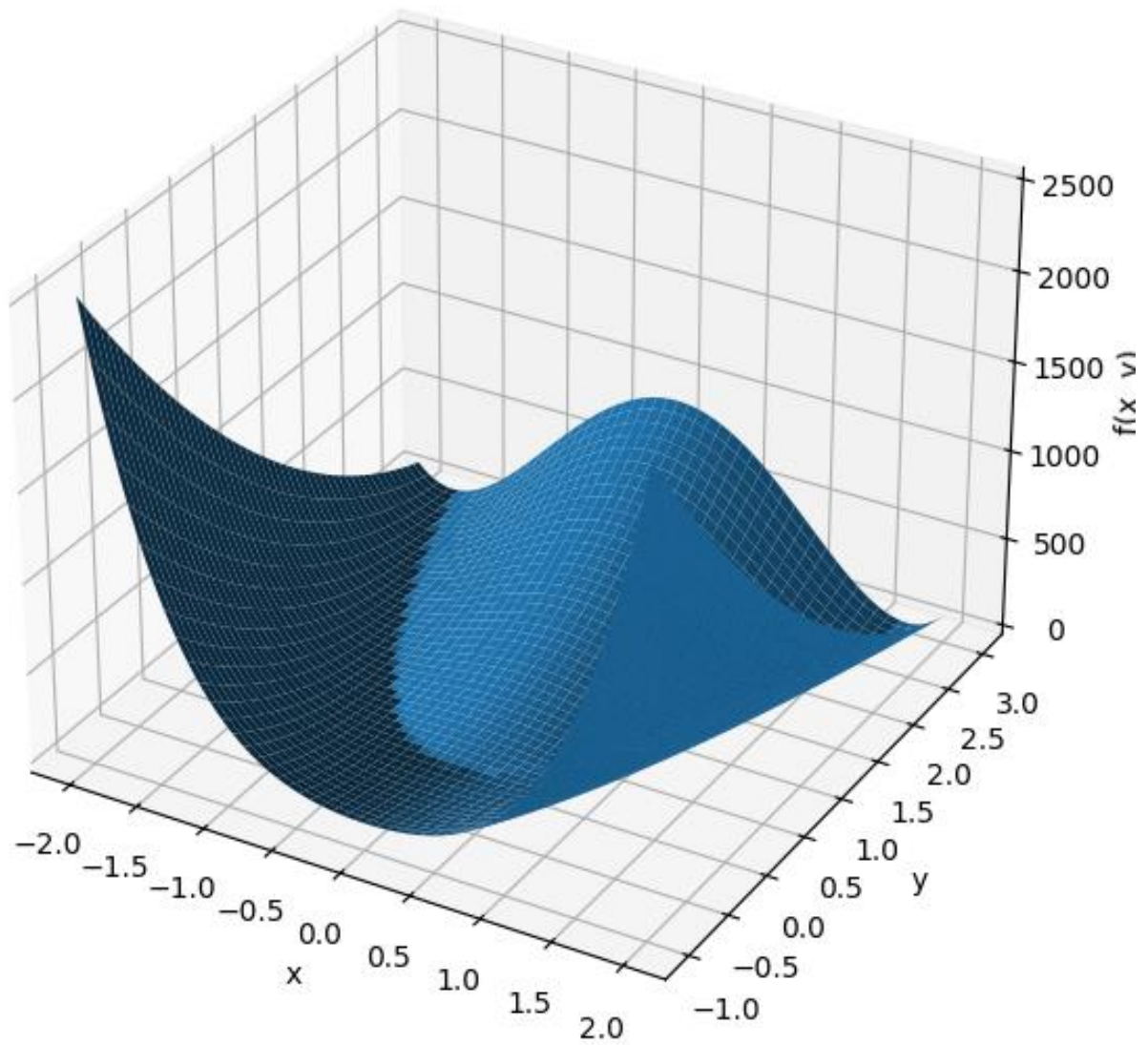


Figure 1. 3D surface plot of the Rosenbrock function from the notebook.

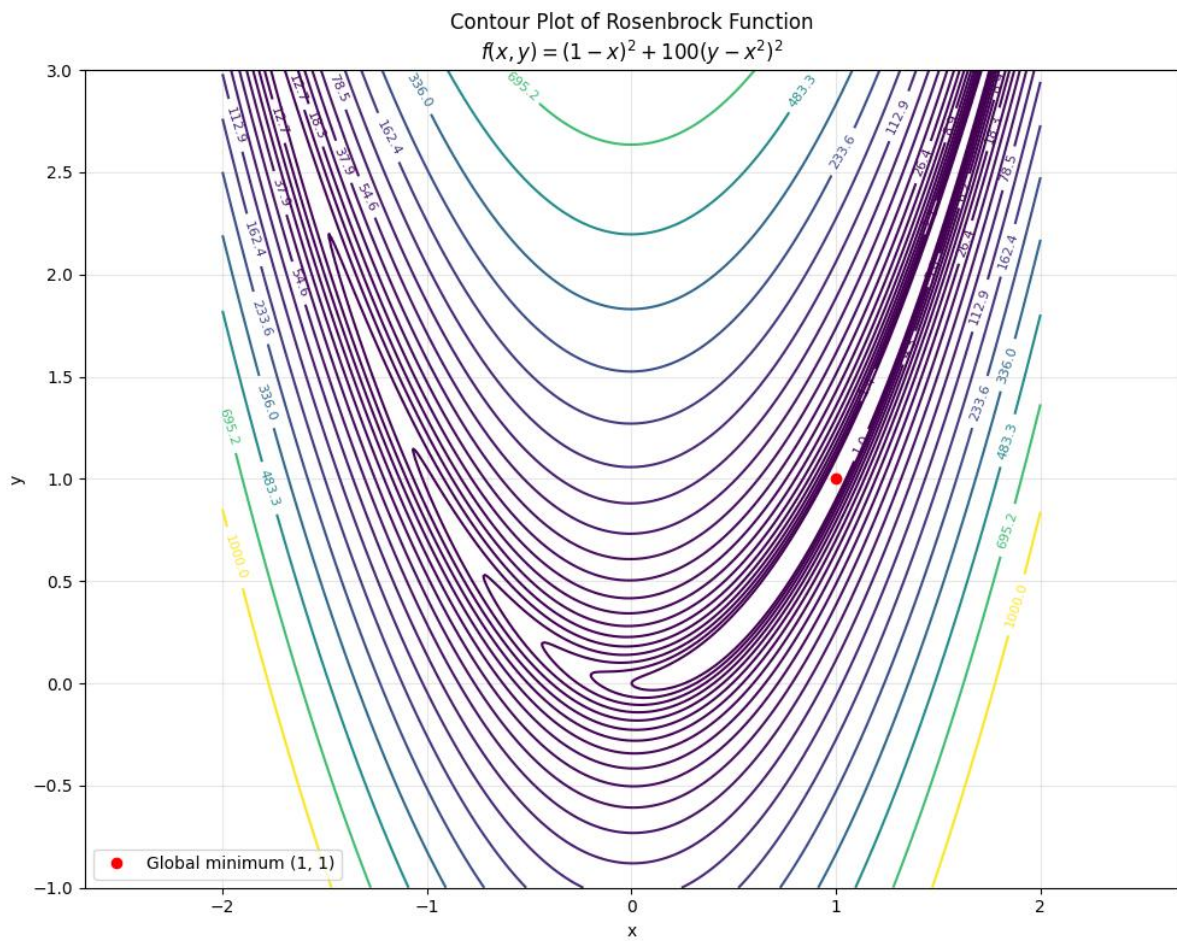


Figure 2. Contour plot of the Rosenbrock function. The marked point (1, 1) is the global minimum.

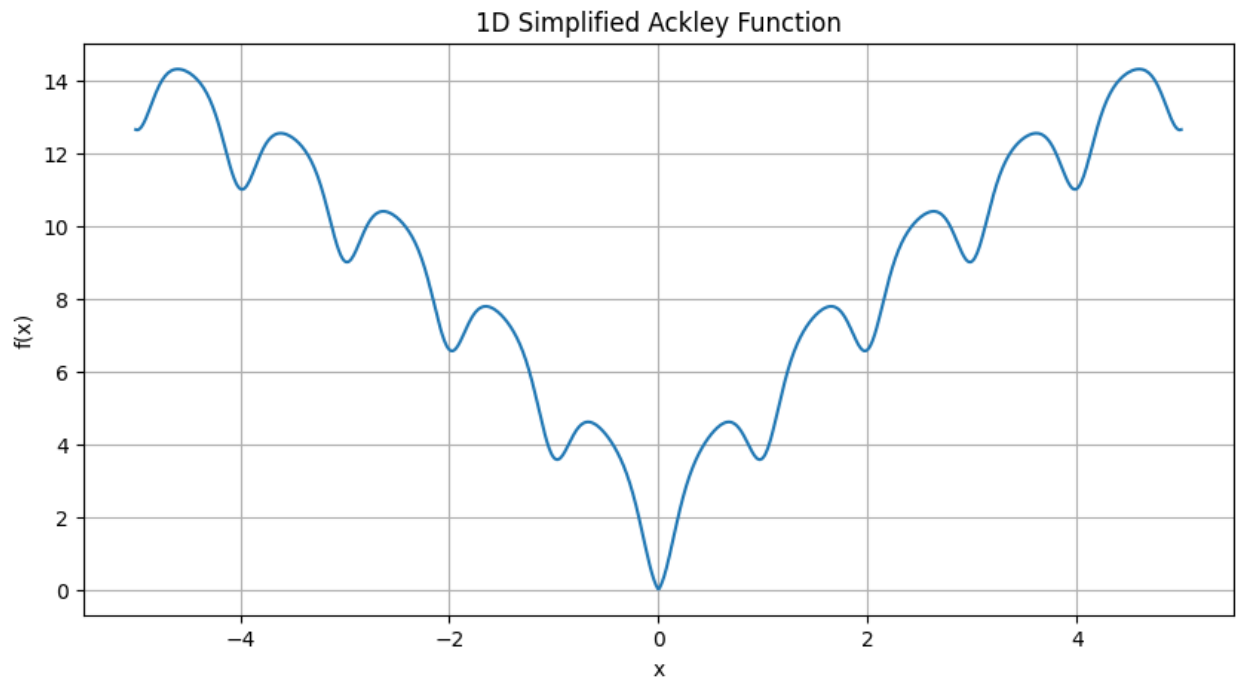
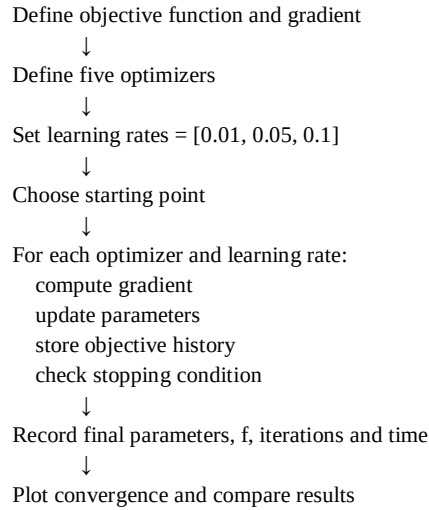


Figure 3. 1D simplified Ackley function from the notebook.

4. Code Flow

The notebook follows a simple repeated flow. The same optimizer functions are used for both objective functions, so the experiment stays consistent.



4.1 Stopping Condition

The notebook used a maximum of 10,000 iterations and stopped early when the change in the parameter vector became smaller than 10^{-8} . Therefore, 'Converged' in the tables means that the parameter-change condition was reached before the iteration limit. A run that stayed finite but did not reach this condition was reported as 'Hit max_iter (not converged)'. Runs that produced non-finite values were reported as 'Diverged'.

4.2 Optimizer Settings

Optimizer	Main update idea	Settings used
Gradient Descent	$x \leftarrow x - \alpha g$	No extra parameter
Momentum	Uses accumulated velocity	$\beta = 0.9$
RMSprop	Scales gradient using recent squared gradients	$\beta = 0.9, \epsilon = 10^{-8}$
Adagrad	Uses accumulated squared gradients	$\epsilon = 10^{-8}$
Adam	Momentum + adaptive scaling	$\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$

5. Experimental Setup

Item	Value
Functions	Rosenbrock; simplified 1D Ackley
Optimizers	Gradient Descent, Momentum, RMSprop, Adagrad, Adam
Learning rates	0.01, 0.05, 0.1
Maximum iterations	10,000
Stopping tolerance	10^{-8} on parameter change
Main Rosenbrock start	(-1.5, 2.0)
Main Ackley start	2.3
Measured	Final point, objective value, iterations, execution time, status

6. Results – Rosenbrock Function

The Rosenbrock experiments show a clear difference between the optimizers. Gradient Descent and Momentum became numerically unstable at all three learning rates. RMSprop and Adagrad remained finite but did not satisfy the selected stopping condition within 10,000 iterations. Adam was the only optimizer that converged for all three learning rates.

Optimizer	LR	Final point	Final f	Iterations	Time (s)	Status
GD	0.01	NaN, NaN	NaN	10,000	0.2007	Diverged
Momentum	0.01	NaN, NaN	NaN	10,000	0.2190	Diverged
RMSprop	0.01	(0.980143, 0.975531)	2.2451×10^{-2}	10,000	0.1805	Not converged
Adagrad	0.01	(-1.071888, 1.155444)	4.2969	10,000	0.1684	Not converged
Adam	0.01	(0.999999, 0.999999)	2.6237×10^{-13}	4,520	0.1134	Converged
GD	0.05	NaN, NaN	NaN	10,000	0.1823	Diverged
Momentum	0.05	NaN, NaN	NaN	10,000	0.2239	Diverged
RMSprop	0.05	(0.725000, 0.465625)	4.3563×10^{-1}	10,000	0.1642	Not converged
Adagrad	0.05	(0.895422, 0.801455)	1.0947×10^{-2}	10,000	0.1460	Not converged
Adam	0.05	(1.000000, 0.999999)	1.0021×10^{-13}	2,105	0.0466	Converged
GD	0.1	NaN, NaN	NaN	10,000	0.1768	Diverged
Momentum	0.1	NaN, NaN	NaN	10,000	0.1983	Diverged
RMSprop	0.1	(0.300000, 0.015000)	1.0525	10,000	0.1663	Not converged
Adagrad	0.1	(0.989299, 0.978680)	1.1461×10^{-4}	10,000	0.1439	Not converged
Adam	0.1	(1.000000, 1.000000)	5.8629×10^{-14}	1,541	0.0353	Converged

Best main Rosenbrock result: Adam with learning rate 0.1 reached (1.000000, 1.000000) with $f(x,y) = 5.8629 \times 10^{-14}$ in 1,541 iterations and about 0.0353 s. This is effectively the global minimum $f = 0$.

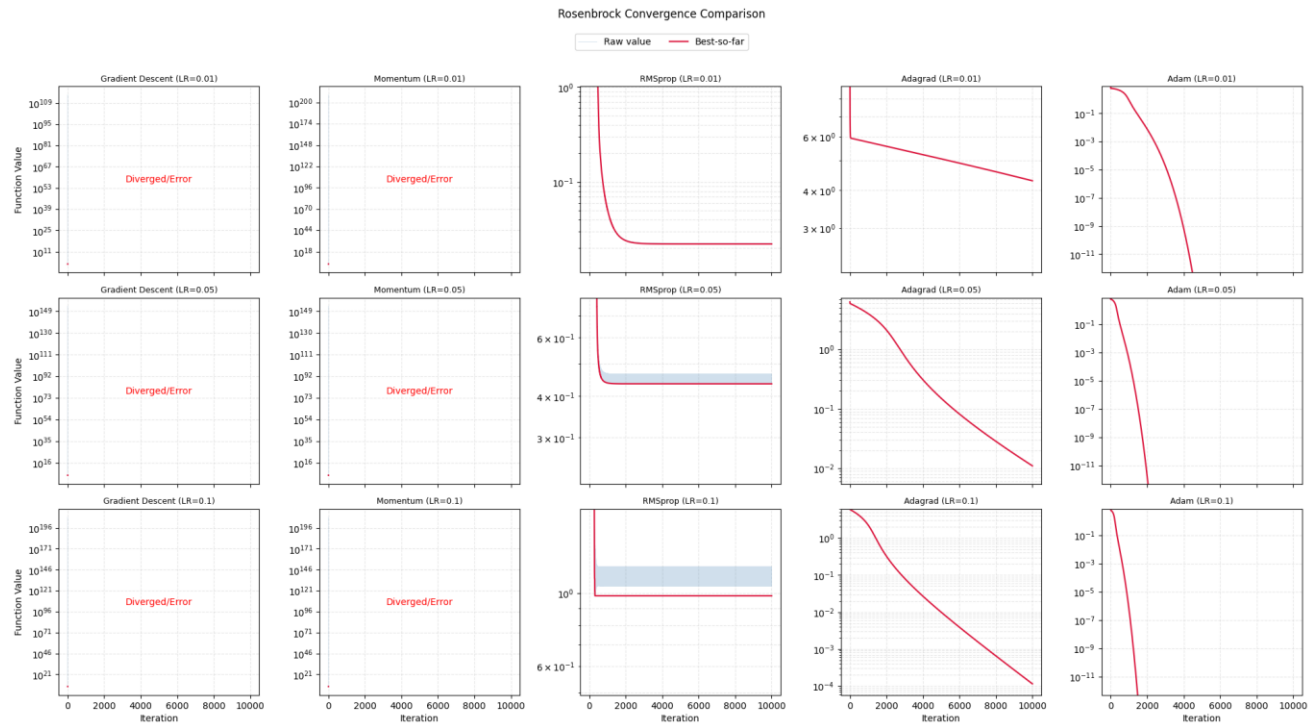


Figure 4. Rosenbrock convergence comparison from the notebook. The plots show the best-so-far objective value on a logarithmic scale.

7. Effect of Learning Rate on Rosenbrock

- The learning-rate experiment followed the expected pattern, but the effect depended strongly on the optimizer. For basic Gradient Descent and Momentum, all three learning rates were too aggressive for the narrow curved Rosenbrock valley and the runs diverged. For Adam, increasing the learning rate clearly improved the result: the number of iterations fell from 4,520 at 0.01 to 2,105 at 0.05 and 1,541 at 0.1, while the final objective remained extremely close to zero.

- Adagrad also improved as the learning rate increased. Its final objective changed from about 4.30 at 0.01 to 0.0109 at 0.05 and 0.000115 at 0.1. RMSprop showed the opposite trend in this experiment: its final objective became worse as the learning rate increased. Therefore, a higher learning rate was not automatically better; it worked well with Adam here because Adam's adaptive scaling helped control the effective step size.

Optimizer	LR = 0.01	LR = 0.05	LR = 0.1	Observed trend
Gradient Descent	Diverged	Diverged	Diverged	Unstable at all tested LRs
Momentum	Diverged	Diverged	Diverged	Unstable at all tested LRs
RMSprop	0.0225	0.4356	1.0525	Higher LR worsened final value
Adagrad	4.2969	0.0109	0.000115	Higher LR improved result
Adam	2.62×10^{-13}	1.00×10^{-13}	5.86×10^{-14}	Higher LR was faster and remained stable

8. Do the Experimental Results Match Optimizer Theory?

Optimizer	Expected behaviour	Observed result	Match?
Gradient Descent	Simple and sensitive to the learning rate; can be unstable on steep/narrow valleys.	Diverged for all three learning rates on Rosenbrock.	Yes
Momentum	Can accelerate movement but may overshoot when gradients are large or steps are too large.	Diverged for all three learning rates.	Yes
RMSprop	Adaptive scaling should help with uneven gradient magnitudes.	Stayed finite, but convergence was slow and final value depended strongly on LR.	Mostly
Adagrad	Adaptive steps; accumulated gradients can make later updates smaller.	Stable but slow on Rosenbrock; higher LR improved the final value.	Mostly
Adam	Combines momentum and adaptive scaling and is usually more robust to the choice of LR.	Converged for all three LRs and reached $f \approx 0$.	Yes

Overall, the Rosenbrock results agree well with the usual theory. The main point that stood out was the advantage of adaptive methods on a function with a narrow curved valley. Adam was able to move through this region much more reliably than plain Gradient Descent or Momentum.

9. Results – Simplified Ackley Function

The Ackley results are different from Rosenbrock because the function has several local minima. The main experiment started at $x = 2.3$. At learning rate 0.01, all five optimizers reached a stopping point, but most ended near $x = 1.97445$ with $f(x) \approx 6.559645$. Momentum instead reached $x \approx -0.968478$ with $f(x) \approx 3.574452$. Neither value is the global minimum.

Optimizer	LR	Final x	Final f	Iterations	Time (s)	Status
GD	0.01	1.974452	6.559645	8	0.0010	Converged
Momentum	0.01	-0.968478	3.574452	273	0.0102	Converged
RMSprop	0.01	1.974452	6.559645	56	0.0052	Converged
Adagrad	0.01	1.974453	6.559645	1,696	0.0600	Converged
Adam	0.01	1.974451	6.559645	257	0.0102	Converged
GD	0.05	-0.246168	2.654742	10,000	0.3854	Not converged
Momentum	0.05	-30.883541	20.572361	10,000	0.3036	Not converged
RMSprop	0.05	1.998115	6.588734	10,000	0.3510	Not converged
Adagrad	0.05	1.974452	6.559645	94	0.0039	Converged
Adam	0.05	1.974451	6.559645	230	0.0089	Converged
GD	0.1	0.102563	0.900049	10,000	0.2719	Not converged
Momentum	0.1	1344.237195	21.634598	10,000	0.3390	Not converged
RMSprop	0.1	2.018482	6.661309	10,000	0.3329	Not converged
Adagrad	0.1	1.974452	6.559645	25	0.0009	Converged
Adam	0.1	1.974457	6.559645	206	0.0081	Converged

Best main Ackley result: Momentum at learning rate 0.01 obtained the lowest objective among the converged runs from the main starting point $x = 2.3$, with $f(x) = 3.574452$ at $x \approx -0.968478$. This should not be called the global optimum; the known global objective is 0 at $x = 0$.

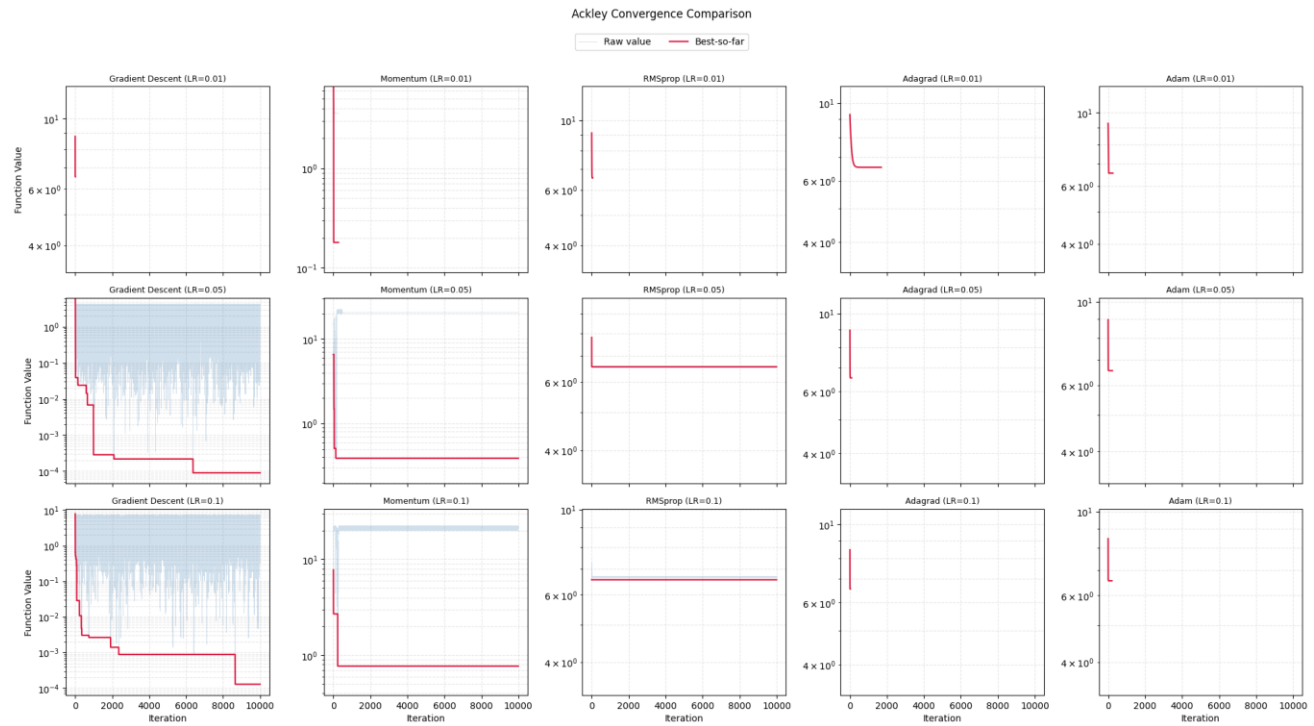


Figure 5. Simplified Ackley convergence comparison from the notebook.

10. Effect of Learning Rate on Ackley

- For Ackley, the effect of learning rate was more mixed. At 0.01, every optimizer satisfied the stopping condition, although they settled at local minima. At 0.05 and 0.1, Gradient Descent, Momentum and RMSprop generally did not satisfy the stopping condition within 10,000 iterations. Momentum became especially unstable at the larger learning rates, ending at very large positive or negative x values.

- Adagrad and Adam were more stable. Adagrad became much faster as the learning rate increased: 1,696 iterations at 0.01, 94 at 0.05 and only 25 at 0.1, while the final objective stayed at about 6.559645. Adam also stayed stable and reached the same local minimum in 257, 230 and 206 iterations respectively.

11. Theory vs Experimental Results on Ackley

The Ackley experiment also gives a good match with optimizer theory, but with an important non-convex effect. Adaptive methods such as Adam and Adagrad were less sensitive to the larger learning rates than basic Gradient Descent and Momentum. This is consistent with their adaptive step-size mechanisms.

At the same time, the experiment shows a limitation of all gradient-based optimizers: reaching a local minimum is possible even when the global minimum is known to be elsewhere. Therefore, the result 'converged' only means that the chosen stopping condition was reached, not that the global minimum was guaranteed.

12. Additional Check: Effect of Starting Point

An additional experiment was carried out by changing the starting point. This was not needed for the main 30 runs, but it helps explain the Ackley results. The best converged objective for each starting point was:

Starting point	Best converged $f(x)$
-2.3	3.574452
0.0	0.000000
0.7	3.574452
1.5	3.574452
2.0	3.574452
2.3	3.574452
3.5	3.574452

This strongly shows the role of initialization. When the starting point was exactly $x = 0$, the global objective of 0 was obtained immediately. For the other tested starting points, the best converged result was generally the local minimum with $f(x) \approx 3.574452$.

13. How Much of the Objective Was Achieved?

Function	Global optimum	Best main-run result	Conclusion
Rosenbrock	$f(1,1) = 0$	5.8629×10^{-14} with Adam, LR 0.1	Global optimum effectively reached
Ackley	$f(0) = 0$	3.574452 from $x = 2.3$	Global optimum not reached in main run

For Rosenbrock, the objective was essentially achieved because the final value was extremely close to zero and the parameters were (1,1). For Ackley, the main experiment did not reach zero. The lowest main-run objective was 3.574452, so the optimizer had converged to a local minimum rather than the global minimum.

14. Observations and Limitations

- The convergence test was based on parameter change, not directly on the gradient norm. Therefore, a 'converged' label means the parameter updates became very small.
- A run reaching 10,000 iterations is described as not satisfying the selected stopping condition; it is not automatically treated as a complete failure.
- The Ackley function is non-convex, so the final solution depends strongly on the starting point.
- The timing values are small and can vary slightly between runs depending on the machine and notebook environment.
- The experiment used the same learning-rate choices for all optimizers, even though different optimizers can have different useful learning-rate ranges.

15. Conclusion

The experiments show that optimizer performance depends on both the function and the learning rate. On the Rosenbrock function, Adam gave the most consistent result and reached the global minimum for all three tested learning rates. Plain Gradient Descent and Momentum were unstable, while RMSprop and Adagrad were stable but slower under the selected stopping condition.

On the simplified Ackley function, the main difficulty was not only optimization speed but also the presence of local minima. Most methods converged to a non-global solution from $x = 2.3$. Adagrad and Adam handled the larger learning rates better than the basic methods, but neither guaranteed the global optimum. Overall, the experimental results mostly followed the expected theory: larger learning rates can make convergence faster when the optimizer is stable, but can also cause overshooting or instability. Adaptive optimizers were more robust in these experiments.